FIITJEE

# JEE(Advanced)-2018

ALL INDIA TEST SERIES

## ANSWERS, HINTS & SOLUTIONS

## FULL TEST – IV
## (Paper-1)

| Q. No. | PHYSICS | Q. No. | CHEMISTRY | Q. No. | MATHEMATICS |
|---|---|---|---|---|---|
| 1. | **A, C, D** | 21. | **A, B** | 41. | **B, D** |
| 2. | **A, B, D** | 22. | **A, B, C** | 42. | **A, C** |
| 3. | **A, D** | 23. | **A, C, D** | 43. | **A, C** |
| 4. | **A, B, C** | 24. | **B, C** | 44. | **A, B, D** |
| 5. | **A, B, C, D** | 25. | **A, D** | 45. | **A, D** |
| 6. | **A, B, C, D** | 26. | **B** | 46. | **A, B, D** |
| 7. | **A, B, D** | 27. | **A, B, D** | 47. | **A, B** |
| 8. | **A, B, C, D** | 28. | **A, B, C** | 48. | **B, C** |
| 9. | **A, C, D** | 29. | **A, B, C** | 49. | **A, B** |
| 10. | **A, D** | 30. | **A, C** | 50. | **A, B, C** |
| 11. | **(A) – (p, r, s, t);<br>(B) – (p, s, t);<br>(C) – (q, r);<br>(D) – (r, s, t)** | 31. | **(A) – (q);<br>(B) – (p, r,s);<br>(C) – (q);<br>(D) – (s)** | 51. | **(A) – (q);<br>(B) – (p);<br>(C) – (r);<br>(D) – (s)** |
| 12. | **(A) – (r, s);<br>(B) – (q, s);<br>(C) – (p, t);<br>(D) – (p, s, t)** | 32. | **(A) – (s);<br>(B) – (q);<br>(C) – (r);<br>(D) – (t)** | 52. | **(A) – (q),<br>(B) – (p),<br>(C) – (p),<br>(D) - (t)** |
| 13. | **5** | 33. | **5** | 53. | **3** |
| 14. | **5** | 34. | **4** | 54. | **8** |
| 15. | **1** | 35. | **4** | 55. | **3** |
| 16. | **3** | 36. | **5** | 56. | **3** |
| 17. | **7** | 37. | **5** | 57. | **2** |
| 18. | **3** | 38. | **6** | 58. | **4** |
| 19. | **6** | 39. | **8** | 59. | **2** |
| 20. | **4** | 40. | **6** | 60. | **1** |

# Physics

## PART – I

### SECTION – A

1. **A, C, D**

$V_1 = V_2 \Rightarrow x_L = x_C$

$\Rightarrow f = \dfrac{1}{2\pi\sqrt{LC}} = 125$ Hz

$I_0 = \dfrac{v_0}{R} = \dfrac{200}{100} \qquad (\because x = 0,\ \therefore z = R)$

$= 2$ A

$V_1 = V_2 = IX_L = I.(\omega L) = I.L.\dfrac{1}{\sqrt{LC}} = I\sqrt{\dfrac{L}{C}} = 2\sqrt{\dfrac{(2/\pi)}{(8\times10^{-6}/\pi)}}$ = 1000 volt

2. **A, B, D**

For Balmer series, $n_1 = 2$ $n_2 = 3, 4, \ldots\ldots$
(lower) (higher)

$\therefore$ In transition (VI), photon of Balmer series is absorbed.

In transition II

$E_2 = -3.4$ eV, $E_4 = -0.85$ eV

$\Delta E = 2.55$ eV

$\Delta E = \dfrac{hc}{\lambda} \Rightarrow \lambda = \dfrac{hc}{\Delta E}$

$\lambda = 486$ nm

Wavelength of radiation = 103 nm = 1030 $\overset{o}{A}$

$\therefore \Delta E = \dfrac{12400}{1030} \simeq 12.0$ eV

So difference of energy should be 12.0 eV (approx.)

Hence $n_1 = 1$ and $n_2 = 3$

(–13.6) eV (–1.51) eV

$\therefore$ Transition is V.

For longest wavelength, energy difference should be minimum.

So, in visible portion of hydrogen atom, minimum energy emitted is in transition IV.

3. **A, D**

(i) From Moseley's law $\sqrt{\upsilon} = a(z-b)$

$\upsilon = 0$ at z = b. From graph, $\upsilon = 0$ at z = 1

$\therefore b = 1$ $\therefore$ $K_\alpha$ Ans.

(ii) For $K_\alpha$, $\sqrt{\upsilon} = \sqrt{\dfrac{3RC}{4}}(z-1)$

Put z = 101 $\Rightarrow \sqrt{\upsilon} = \sqrt{\dfrac{3RC}{4}}.100$

Photon energy = $h\upsilon = \dfrac{3}{4}hRC.100^2 = \dfrac{3}{4}\times13.6\times100^2$ = 102 keV

4. **A, B, C**

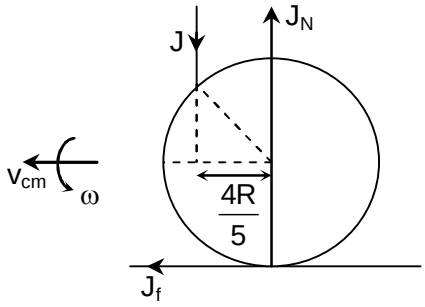
$J_N = J \qquad J_f = mv_{cm}$

$$J\frac{4R}{5} - J_f R = \frac{2}{5}mR^2\omega$$

$$\frac{4}{5}J - J_f = \frac{2}{5}mv_{cm} = \frac{2}{5}J_f$$

$$\Rightarrow \frac{4}{5}J = \frac{7}{5}J_f \Rightarrow J_f = \frac{4}{7}J$$

$$\mu J > \frac{4}{7}J \Rightarrow \mu > \frac{4}{7} \text{ for rolling}$$




$J_f \le \mu J_N$

5. **A, B, C, D**

$$\frac{1}{2}mv_0^2 = \frac{5mgL}{2} = K + mgL(1-\cos\theta)$$

$$\Rightarrow K = 2mgL\cos^2\frac{\theta}{2} + \frac{mgL}{2}$$

$$T = 6mg\cos^2\frac{\theta}{2}$$

$$a_r = g + 4g\cos^2\frac{\theta}{2}$$

$$a_t = g\sin\theta = g\sqrt{1-\cos^2\theta}$$

$$\Rightarrow a_t^2 = g^2 - g^2\cos^2\theta$$

6. **A, B, C, D**

$$mg = \frac{KQqx}{\left(y^2+x^2\right)^{\frac{3}{2}}} \Rightarrow \frac{q}{m} = \frac{4\pi\varepsilon_0\left(y^2+x^2\right)^{\frac{3}{2}}g}{Qy}$$

7. **A, B, D**

$$\rho gAh = \left(\sigma_1 Ah_1 + \sigma_2 Ah_2\right)g$$

$$\Rightarrow \sigma_1 h_1 + \sigma_2 h_2 = \rho h$$

$$\sigma_1 h_1 + \sigma_2 h_2 = \left(h_1 + h_2\right)\rho \qquad \left(\because h_1 + h_2 = h\right)$$

$$\Rightarrow \frac{h_2}{h_1} = \frac{\sigma_1 - \rho}{\rho - \sigma_2}$$

If the cylinder is displace down by y

$$F_{net} = \rho Ahg - \left(\sigma_1 A\left(h_1 - y\right) + \sigma_2 A\left(h_2 + y\right)\right)g$$

$$= \rho Ahg - \left(\sigma_1 h_1 A + \sigma_2 h_2 A\right)g + \left(\sigma_1 - \sigma_2\right)Agy = \left(\sigma_1 - \sigma_2\right)Agy$$

$$W = \left(\sigma_1 - \sigma_2\right)Ag\frac{h_1^2}{2} = \frac{1}{2}\rho Ahv^2$$

$$\Rightarrow v = \sqrt{\left|\frac{\sigma_1 - \sigma_2}{\rho}\right|\frac{gh_1^2}{h}}$$

8. **A, B, C, D**

$\lambda = 4.9 \times 10^{-18}\ s^{-1}$

$\tau_{avg} = \frac{1}{\lambda} = 6.49 \times 10^9$ years

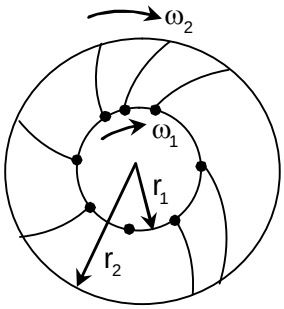
$t_{1/2} = \frac{0.693}{\lambda} = 4.5 \times 10^9$ years

No. of nuclei's remaining after $9 \times 10^9$ hrs is $\frac{N_0}{4}$

$\Rightarrow$ activity reduces to $\frac{1}{4}$

After 3 half lives nuclei decayed is 87.5%.

9. **A, C, D**




As the sand leaves the inner drum through the open holes, it does not exert any force of the drum, angular momentum remains conserved.

At time t,

$$m_1 r_1^2 \omega_0 = (m_1 - \lambda t) r_1^2 \omega_0 + (m_2 + \lambda t) r_2^2 \omega_2$$

$$\omega_2 = \frac{\lambda t r_1^2 \omega_0}{(m_2 + \lambda t) r_2^2}$$

The speed of inner drum does not change.

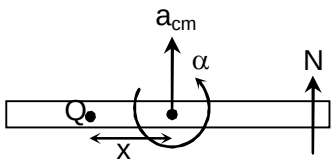
10. **A, D**

In first case, we can write torque equation about bottom point because it coincides with instantaneous axis of rotation while in second case, you will write torque equation of centre of mass.

## SECTION – B

11. **(A) – (p, r, s, t); (B) – (p, s, t); (C) – (q, r); (D) – (r, s, t)**

During collision:




$a_{cm} = \frac{N}{m}$

$\frac{m^2}{12}\alpha = \frac{NL}{2} \qquad \alpha = \frac{6N}{L}$

Acceleration of point Q is $a_Q = a_{cm} - \frac{L}{6}\alpha = 0$

Hence, angular momentum will be conserved about point Q.
During collision:
Torque about point N on rod
$\tau_{ext.} + \tau_{pseudo\ force} = 0$
Hence, angular momentum will be conserved about point N also.

12. **(A) – (r, s); (B) – (q, s); (C) – (p, t); (D) – (p, s, t)**

## SECTION – C

13. **5**

$1\times\frac{1}{2}\times10+1\times80+1\times1\times T_{eq} = 0.1\times\frac{1}{2}\times100+0.1\times540+0.1\times1(100-T_{eq})$

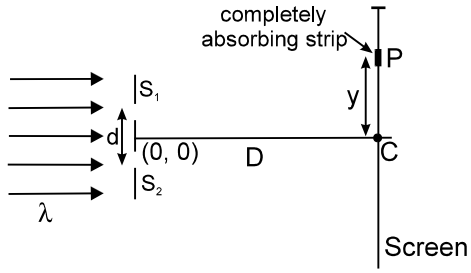
$T_{eq} = \frac{290}{11}K = \frac{58}{11}\times5$

14. **5**

$F = \sum\eta A\frac{dv}{dr} = 0.5\times0.75\left[\frac{0.5}{10\times10^{-3}}+\frac{0.5}{15\times10^{-3}}\right] = 31.25$

15. **1**

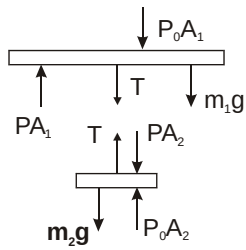
Let $I_0$ be the intensity at P due to individual slits $S_1$ or $S_2$.
When waves due to both the sources superpose at P, the net intensity at P is $I = 4I_0\cos^2\frac{\pi y}{\beta}$, where $\beta = \frac{\lambda D}{d}$
Since force on strip at P is twice due to individual source
Hence $I = 2\ I_0$




$\therefore \cos^2\frac{\pi y}{\beta} = \frac{1}{2}$ or $\frac{\pi y}{\beta} = \frac{\pi}{4}$ or $y = \frac{\beta}{4} = \frac{\lambda D}{4d}$

16. **3**

Due to the heating pressure inside is not changed. Let inside pressure be $\rho$. Then for equilibrium of the system:




$P(A_1 - A_2) = P_0(A_1 - A_2) + (m_1 + m_2)g$

$\Rightarrow P\Delta V = (P_0\Delta A + mg)L$

L is displacement of the piston.

$P.\Delta V = nR\Delta T$

$\Delta T = \frac{P\Delta V}{nR} = \frac{(P_0\Delta V + mg)L}{nR}$

$= \frac{(10^5 P_a\times10^{-3}m^2 + 5\times10)(50\times10^{-2})}{1\times R} = \frac{75}{R}\ K$

17. **7**

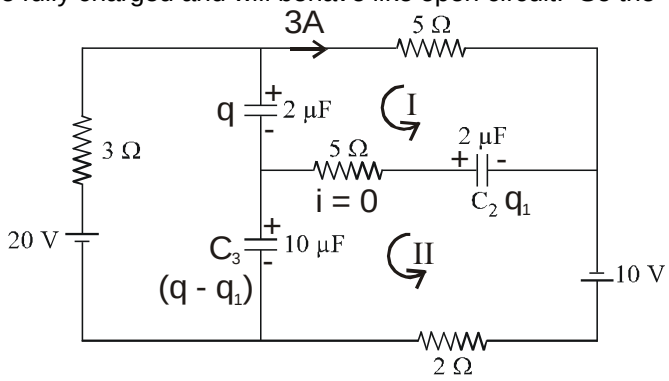
$$mgH_B = \frac{1}{2}mv^2$$

$$mgH_S = \frac{1}{2}mv^2 + \frac{1}{2}I\omega^2 = \frac{1}{2}mv^2 + \frac{1}{2}\times\frac{1}{2}mr^2\omega^2 = \frac{1}{2}\times\frac{7}{5}mv^2$$

$$\frac{H_S}{H_B} = \frac{7}{5} \Rightarrow \frac{5H_S}{H_B} = 7$$

18. **3**

During the steady state the capacitors are fully charged and will behave like open circuit. So the current will flow outer loop only. The current in the outer loop will be $= \frac{30}{10}$ = 3 Amp. in clockwise direction. Charge distribution on the plates of capacitors are as shown in figure.




Applying KVL in loop – I

$$\frac{q}{2} + \frac{q_1}{2} - 15 = 0 \Rightarrow q + q_1 = 30 \quad \text{..... (1)}$$

Applying KVL in loop – II

$$\frac{q - q_1}{10} - 6 + 10 - \frac{q_1}{2} = 0 \qquad \text{..... (2)}$$

On solving equation (1) and (2) we get $q_1 = 10\ \mu C$

19. **6**

Potential difference across wire AB = 5 V

$\therefore$ p.d. across 40 cm of this wire = $\frac{5}{100}\times 40$ = 2 V

$\therefore$ p.d. across 20 cm of wire CD = 2 V

$\therefore$ p.d. across wire CD = $\frac{2}{20}\times 80$ = 8 V

p.d. across $2\ \Omega$ resistor = 2 x 2 = 4 V

$\therefore$ emf of the cell = 12 V

20. **4**

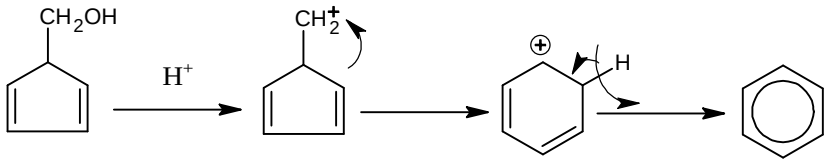
$$\lambda = \lambda_1 + \lambda_2 \Rightarrow \frac{1}{t} = \frac{1}{t_1} + \frac{1}{t_2} \Rightarrow \frac{3}{4} = \frac{1}{2} + \frac{1}{t_2} \Rightarrow t_2 = 4 \text{ years}$$

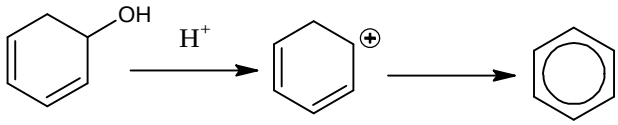
# Chemistry

## PART – II

### SECTION – A

21. **A, B**

(A) 


(B) 

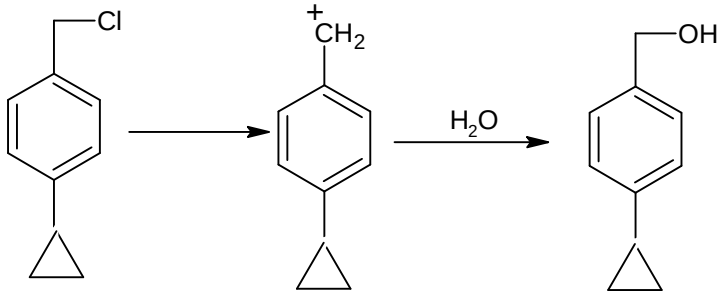

22. **A, B, C**
Alcohols that can form stable carbocation.

23. **A, C, D**
(A) $SOCl_2$ gives retention in configuration
(B) Hoffmann product
(C) Hoffmann product
(D) $E_1$ CB

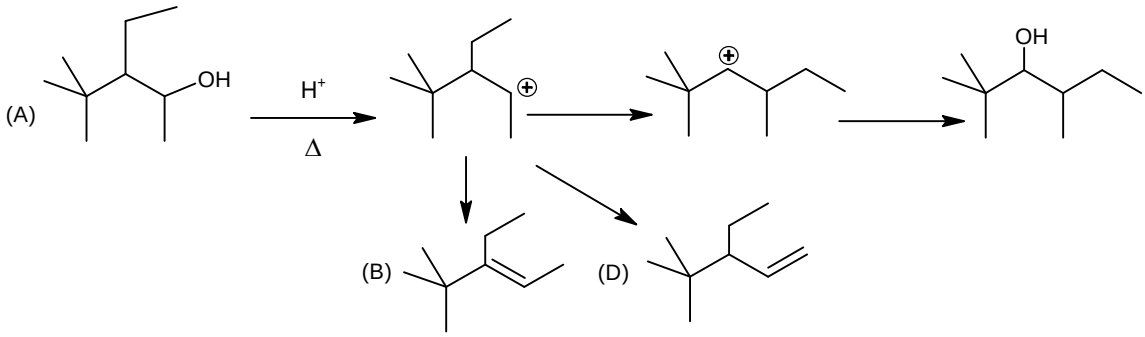
24. **B, C**
(A) Alkene should form
(B) More substituted product
(C) less substituted product
(D) More substituted product

25. **A, D**
(A) Most stable cation
(D) More substituted alkene final product.

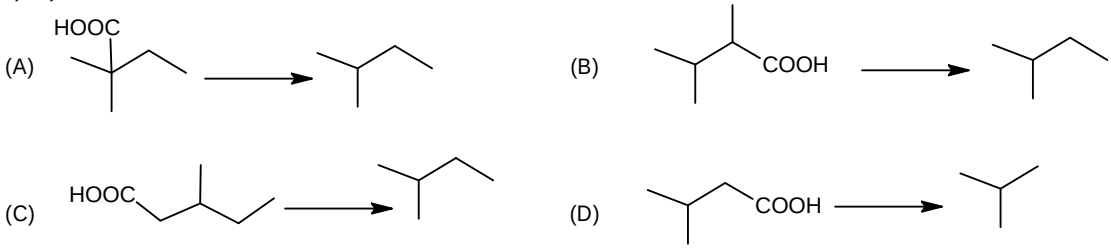
26. **B**




27. **A, B, D**




28. **A, B, C**




29. **A, B, C**

$$NaHCO_3 \xrightarrow{H^+} CO_2$$

$$Na_2CO_3 \xrightarrow{H^+} CO_2$$

$$Na_2SO_3 \xrightarrow{H^+} SO_2$$

30. **A, C**
Fact.

## SECTION – B

31. **(A) – (q); (B) – (p, r, s); (C) – (q); (D) – (s)**
Fact

32. **(A) – (s); (B) – (q); (C) – (r); (D) – (t)**
A - Ea is significant & intermediate is also stable.
B – The Ea of II$^{nd}$ step is more than first step.
C - Ea of II$^{nd}$ step is very small so intermediate will convert into product immediately.
D – There are three TS so it must be three step reaction.

## SECTION – C

33. **5**
Unit cell is fcc with half of the tertreahedical voids are occupied by C – atoms hence effective no is $= 1 + 2 \times \frac{1}{8} = \frac{5}{4}$

34. **4**

$SO_2Cl_2 + H_2O \longrightarrow H_2SO_4 + 2HCl \xrightarrow{4NaOH} Na_2SO_4 + 2NaCl$

35. **4**

CNS, CN, CNO, $NO_2$

36. **5**

Present as $Fe^+$

37. **5**

$$\begin{array}{ccc} N_2 + & 3H_2 & \longrightarrow 2NH_3 \\ 1 & a & 0 \\ 0 & a-3 & 2 \end{array}$$

$$\frac{2}{a-1} = \frac{1}{2}$$

$$4 = a - 1$$

$$a = 5$$

38. **6**

$i = 1.25$

Original mole fraction $= \frac{1}{n} = \frac{1}{1+(n-1)} = \frac{1.25}{1.25+(n-1)} = \frac{1}{5} \Rightarrow n = 6$

39. **8**

$MnO_2 + 4HCl \longrightarrow MnCl_2 + Cl_2 + 2H_2O$

100 gm sol contains 4 gm HCl

$\frac{100}{1.2}$ ml sol contains 4 gm HCl

$nCl_2 = \frac{1.8}{22.4} = 0.0805$

$W_{MnO_2} = 0.0805 \times 87 = 8$ gm

40. **6**

In colemanite every boron carry one OH groups as in borane.

# Mathematics

## PART – III

### SECTION – A

41. **B, D**

$f(x_1+x_2)+f(x_1-x_2)=2f(x_1).f(x_2)\forall x_1,x_2$

on putting $x_1=0,x_2=0$ in above equation $f(0)+f(0)=2(f(0))^2$

$f(0)=1$

on putting $x_1=0,x_2=x_1$ in eqation $(i)$ we get $f(x_1)+f(-x_1)=2f(0)f(x_1)$

$f(x_1)=f(-x_1)$ .....(ii)

$f(5)=f(-5)$

Differentiating equation $(ii)$ w.r.t $x_1$ we get $f'(x_1)+f'(-x_1)=0\Rightarrow f'(5)+f'(-5)=0$

42. **A, C**

$\frac{a_n}{a_{n-1}}=\frac{a_{n-1}}{a_{n-2}}$

Here $a_1,a_2,a_3$.....are in G.P

Let $a_2=x$ then for n = 3

$\frac{a_3}{a_2}=\frac{a_2}{a_1}\Rightarrow a_2^2=a_1a_3\Rightarrow a_3=\frac{x^2}{2}$

$\therefore$ G.P is 2, x, $\frac{x^2}{2},\frac{x^3}{4}$,.....

Common ratio r = $\frac{x}{2}$

Given $\frac{x^4}{8}\le 162\Rightarrow x^4\le 1296\Rightarrow x\le 6$

Also, x & $\frac{x^4}{8}$ are integers

so, if x is even, then only $\frac{x^4}{8}$ will be an integer

Hence, the possible values of x are 4 & 6, because $x\neq 2$ as terms are distinct hence possible values of $a_5=\frac{x^4}{8}$ are $\frac{4^4}{8}$ & $\frac{6^4}{8}$.

43. **A, C**

$\frac{d}{dx}f_n(x)=\frac{d}{dx}\left\{e^{f_{n-1}(x)}\right\}$

$=e^{f_{n-1}(x)}\frac{d}{dx}f_{n-1}(x)$

$=f_n(x)\frac{d}{dx}(f_{n-1}(x))$

$= f_n(x)\frac{d}{dx}\left(e^{f_{n-2}(x)}\right)$

$= f_n(x)e^{f_{n-2}(x)}\frac{d}{dx}f_{n-2}(x)$

$= f_n(x)f_{n-1}(x)\frac{d}{dx}f_{n-2}(x)$

$= f_n(x)f_{n-1}(x).....f_2(x)\frac{d}{dx}f_1(x)$

$= f_n(x)f_{n-1}(x).....f_2(x)e^{f_0(x)}\frac{d}{dx}f_0(x)$

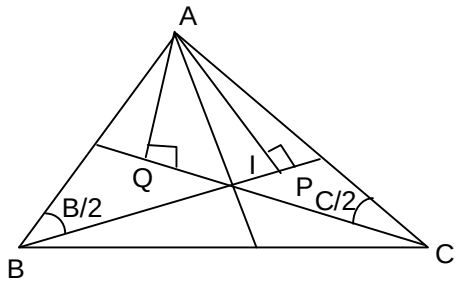
44. **A, B, D**

In $\Delta APB$, we have $AP = AB \sin B/2$

In $\Delta AIB$, we have $\frac{BI}{AB} = \frac{\sin(A/2)}{\cos(C/2)}$

$\Rightarrow \frac{AP}{BI} = \frac{\sin B/2\cos C/2}{\sin A/2}$

Similarly, $\frac{AQ}{CI} = \frac{\sin C/2\cos B/2}{\sin A/2}$




Now, $\frac{AP}{BI} + \frac{AQ}{CI} = \frac{\sin\left(\frac{B+C}{2}\right)}{\sin A/2} = \cot(A/2)$

$\cot(A/2) = \sqrt{3}, \angle A = 60^o$

45. **A, D**

$\hat{u}.\hat{v} + \hat{u}.\hat{w} = \frac{1}{2}$ .....(ii) (taking dot product with $\hat{u},\hat{v}$ & $\hat{a}$)

$\hat{u}.\hat{v} + \hat{v}.\hat{w} = \frac{3}{4}$ .....(iii)

$\hat{a}.\hat{w} = \frac{3}{4}$ .....(iv)

Again taking dot product with $\hat{w}$, we get $\hat{u}.\hat{w} + \hat{v}.\hat{w} + |\hat{w}|^2 = \hat{a}.\hat{w}$

$\hat{u}.\hat{w} + \hat{u}.\hat{w} + 1 = \frac{3}{4}$

$\hat{u}.\hat{w} + \hat{u}.\hat{w} = \frac{-1}{4}$ .....(v)

on. adding (ii),(iii),(v)

$\hat{u}.\hat{v} + \hat{v}.\hat{w} + \hat{u}.\hat{w} = \frac{1}{2}$

on subtrating equation (ii) from (iv)

$\hat{v}.\hat{w} = 0$

46. **A, B, D**

$I_1 = \int_1^e x\left(\frac{1+x}{x}\right)(x+\ln x)^{100}dx$

$$= \frac{e(e+1)^{101}-1}{101} - \frac{1}{101}\int_1^e (x+\ln x)^{101}\,dx$$

$$I_2 = \frac{1}{e}\int_1^e (t+\ln t)\,dt \text{ on taking } e\sin x = t$$

$$\Rightarrow I_1 + \frac{eI_2}{101} = \frac{e(e+1)^{101}-1}{101} \Rightarrow K = 1$$

47. **A, B**

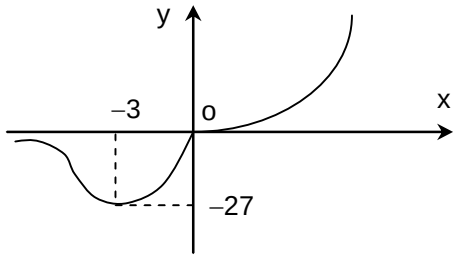
$y = x^3e^{x+3} = -k$

$y = x^3e^{x+3}$

$\frac{dy}{dx} = e^{x+3}\left(3x^2+x^3\right) = 0$

$\Rightarrow x = 0, x = -3$

at $x = -3, y = -27$




For 2 distinct real roots, $k \in (0,27)$

So, prime value of k will be

$2, 3, 5, 7, 11, 13, 17, 19, 23 = 9$

48. **B, C**

Let $2^{\frac{\pi}{\cos^{-1}x}} = t \Rightarrow t \geq 2$

$\therefore$ equation becomes $t^2 - \left(a+\frac{1}{2}\right)t - a^2 = 0$

has one root 2 or greater than 2 & other root less than 2 $\Rightarrow f(2) \leq 0$

$\Rightarrow 4 - \left(a+\frac{1}{2}\right):2 - a^2 \leq 0$

$a^2 + 2a - 3 \geq 0$

$(a+3)(a-1) \geq 0$

$\Rightarrow a \leq -3$ or $a \geq 1$

49. **A, B**

Given circle touches externally at real axis So, the centre C of the described circles lies on real axis which has radius r then $CC_1 = CC_2 = 1+r$

$C_1C_3 = C_2C_3 = 1$

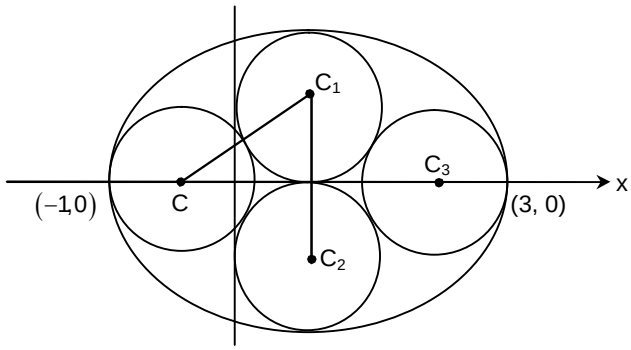
$CC_3 = 2-r$

$\therefore (1+r)^2 = 1^2 + (2-r)^2$

$\Rightarrow r = 2/3$

$\therefore$ Centre of described circle is

$-1+\frac{2}{3} = \frac{-1}{3}$ or $3-\frac{2}{3} = \frac{7}{3}$



$\therefore$ Required equation of circle are

$\left|z+\frac{1}{3}\right|=\frac{2}{3}$ & $\left|z-\frac{7}{3}\right|=\frac{2}{3}$

50. **A, B, C**

$a=(\pi-3)+(\pi-4)+(5-2\pi)=-2$

$f(-2)=f(2)\Rightarrow f(x)$ is many one $\Rightarrow$ non invertible

Let $t = x^2+|x|, t\in[0,\infty)$

$f(x)\in[1,\infty)$

$\Rightarrow b=1 \& a+b=-1$

$$g(x)=\left[(1+\cos 2x)^2-2\cos x-\frac{1}{2}\left(2\cos^2 2x-1\right)-x^7\right]^{1/7}$$

$$g(x)=\left(\frac{3}{2}-x^7\right)^{1/7}$$

$$g(g(x))=\left[\frac{3}{2}-\left(\frac{3}{2}-x^7\right)\right]^{1/7}$$

$f(g(g(b)))=f(b)=e^2$

## SECTION – B

51. **(A) – (q); (B) – (p); (C) – (r); (D) – (s)**

$A(2,2,-1), B(3,1,2) C(1,1,1)$

$\overline{AB}=\hat{i}-\hat{j}+3\hat{k}$

$\overline{AC}=-\hat{i}-\hat{j}+2\hat{k}$

$\overline{BC}=-2\hat{i}-\hat{k}$

$\overline{AC}\perp\overline{BC}$

$\therefore$ D is $(4,2,0)$

equation of base is $x-5y-2z+6=0$

Let E be foot of $\perp$ drawn from P to the base

Equation of EP is $\frac{x-y}{1}=\frac{y+\frac{26}{3}}{-5}=\frac{z+\frac{10}{3}}{-2}=D$

Let coordinates of E be

$\left(4+D,\frac{-26}{3}-5D,\frac{-10}{3}-2D\right)$ since it lies on base

$\therefore 4+D-5\left(\frac{-26}{3}-5D\right)-2\left(\frac{-10}{3}-2D\right)+6=0$ i.e $D=-2$

Coordinates of E are $\left(2,\frac{4}{3},\frac{2}{3}\right)$

$|\overline{EP}|=\sqrt{120}$

Volume $= \frac{1}{3}\left|\overline{AC} \times \overline{BC}\right|$. EP $= 20$ cubic units

52. **(A) – (q), (B) – (p), (C) – (p), (D) - (t)**

$(A) 1-\tan^2 x = \frac{2\tan x}{\tan 2x}$

$$\frac{1-\tan^2 2^k\pi}{2^n+1} = \frac{2\tan\frac{2^k\pi}{2^n+1}}{\tan\frac{2^{k+1}\pi}{2^n+1}}$$

On taking product we get

$(B) 1+2\cos 2a_k = 1+2\left(1-2\sin^2 a_k\right)$

$= 3-4\sin^2 a_k$

$= \frac{\sin 3a_k}{\sin a_k}$

where $a_k = \frac{3^k\pi}{3^n+1}$

$$\frac{\sin 3a_n}{\sin a_1} = \frac{\sin\frac{3^{n+1}\pi}{3^n+1}}{\sin\frac{3\pi}{3^n+1}}$$

$\therefore$ The numeration of last fraction is equal to $\sin\left(3\pi - \frac{3\pi}{3^n+1}\right) = \sin\frac{3\pi}{3^n+1}$

$\therefore$ Ans is 1

$(C) 1-\cos a = \frac{1-4\cos^2 a}{1+2\cos a} = \frac{1-2(1+\cos 2a)}{1+2\cos a}$

$= -\left(\frac{1+2\cos 2a}{1+2\cos a}\right)$

$\therefore$ the product becomes $(-1)^n \frac{1+2\cos x}{1+2\cos\frac{x}{2^n}}$

$(D)\ \tan x = \frac{1}{\tan x} - \frac{1-\tan^2 x}{\tan x} = \frac{1}{\tan x} - \frac{2}{\tan 2x} = \cot x - 2\cot 2x$

$\Rightarrow \frac{1}{2^n}\tan\frac{a}{2^n} = \frac{1}{2^n}\cot\frac{a}{2^n} - \frac{1}{2^{n-1}}\cot\frac{a}{2^{n-1}}$

$\Rightarrow$ on adding $\lim_{n\to\infty} \frac{1}{2^n}\cot\frac{a}{2^n} - \cot a$

$= \frac{1}{a} - \cot a$

$a = 2$

$\Rightarrow$ Ans is $\frac{1}{2}$.

## SECTION – C

53. **3**

$A_i$ = no. of heads obtained by A when tosses two coins
$B_i$ = no. of heads obtained by B when tosses 3 coins

$$P(E) = P\{(A_1 \cap B_0) \cup (A_2 \cap B_2) \cup (A_2 \cap B_1)\}$$

$$= 2C_1\left(\frac{1}{2}\right)\left(\frac{1}{2}\right)\left(\frac{1}{2}\right)^3 + \left(\frac{1}{2}\right)^2\left(\frac{1}{2}\right)^3 + \left(\frac{1}{2}\right)^2 3C_1\left(\frac{1}{2}\right)\left(\frac{1}{2}\right)^2$$

$$= \frac{2}{32} + \frac{1}{32} + \frac{3}{32} = \frac{3}{16}$$

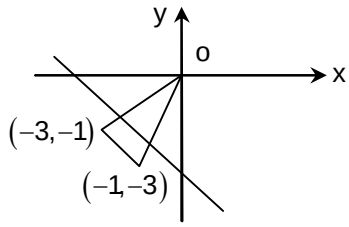
F = {A & B tie a particular game} then $P(F) = P\{(A_0 \cap B_0) \cup (A_1 \cap B_1) \cup (A_2 \cap B_2)\} = \frac{5}{16}$

$P(\text{A finally wins the game})$

$$= P(\text{E or FE or FFE or .....}) = \frac{3}{11}$$

54. **8**

Slope of BC = - 1 slope of line parallel to BC $= -\frac{a}{b}$




$$\therefore \frac{-a}{b} = -1$$

$$\Rightarrow a = b$$

Distance of line $ax + by + 2 = 0$ from the origin

$$= \frac{2}{\sqrt{a^2 + b^2}} = \frac{1}{\sqrt{2}} \Rightarrow a^2 + b^2 = 8$$

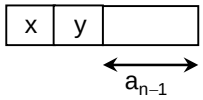
$$\therefore a^2 = b^2 = 4$$

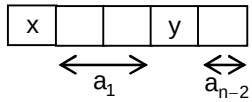
$$\therefore a = b = 2$$

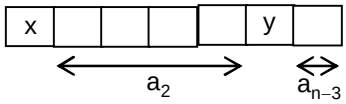
55. **3**

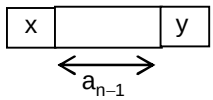
Total arrangements (holding 2 types of coins) $= \frac{8!}{4!4!} = 70$

Let $a_n$ denote the ways of arranging 2 n children out of which n have one Rs.1 coin & rest have Rs.2 coin such that no one waits for change at the counter. Let x denote the child having Rs.1 coins & y denote child having Rs.2 coin in hand.

Case - 1 


Case - 2 


Case - 3 


Case - n 


So, $a_n = a_{n-1} + a_1 a_{n-2} + a_2 a_{n-3} + ..... + a_{n-1}$

$a_1 = 1$

$a_2 = 2$

$a_3 = 2 + 1 + 2 = 5$

$a_4 = 14$

$\therefore$ required probability $(K)= \frac{14}{70}=\frac{1}{5}\Rightarrow 15K=3$

56. **3**

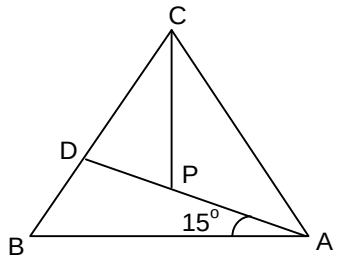
F in segment BC choose point D on segment BC such that $2\,|BD|=|CD|$ & construct ray BP such that $\angle PBC=45^o$ & let A be point on ray BP that moves from B in the direction of the ray. Applying sine rule in $\Delta ACD$ & $\Delta ABC$ Let $\alpha=\angle CAD$ & $\angle CDA=\angle CBA+\angle DAB=60^o$ we have

$\frac{CD}{\sin\alpha}=\frac{CA}{\sin 60^o}$ & $\frac{BC}{\sin(\alpha+15^o)}=\frac{CA}{\sin 45^o}$ Dividing the first equation by second equation gives

$$\frac{|CD|\sin(\alpha+15^o)}{|BC|\sin\alpha}=\frac{\sin 45^o}{\sin 60^o}$$

$$\left|\frac{CD}{BC}\right|=\frac{2}{3}=\left(\frac{\sin 45^o}{\sin 60^o}\right)^2$$

$$\left(\frac{\sin 45^o}{\sin 60^o}\right)^2=\frac{\sin\alpha}{\sin(\alpha+15^o)}.\frac{\sin 45^o}{\sin 60^o}$$




Clearly $\alpha=45^o$ satisfies the equation $\angle ABC=45^o,\angle CAB=60^o$ & $\angle ACB=75^o$

57. **2**

$s=\sin 2x+\sin 4x+.....+\sin 2nx$

$$=\frac{\sin nx-\sin(n+1)x}{\sin x}$$

$c=\cos 2x+\cos 4x+.....+\cos 2nx$

$$=\frac{\sin nx\ \cos(n+1)x}{\sin x}$$

$$\left(\frac{\sin^2 nx}{\sin^2 x}\right)^2=\left(\frac{\sin nx\sin(n+1)x}{\sin x}\right)^2+\left(\frac{\sin nx\cos(n+1)x}{\sin x}\right)^2=s^2+c^2$$

On the other hand $s^2+c^2=(\sin 2x+\sin 4x+.....\sin 2nx)^2+(\cos 2x+\cos 4x+.....+\cos 2nx)^2$

$$=n+\sum_{1\le k\le n}(2\sin 2 x\sin 2kx+2\cos 2 x\cos 2kx)$$

$$=n+2\sum_{1\le k\le n}\cos 2(k-1)x$$

$\Rightarrow a_{1,k}=2$

58. **4**

By Schwartz's inequality $1=\left|\int_0^1 f(x)dx\right|=\left|\int_0^1\frac{f(x)\sqrt{1+x^2}dx}{\sqrt{1+x^2}}\right|$

$$1\le\sqrt{\int_0^1(1+x^2)f^2(x)dx}\sqrt{\int_0^1\frac{dx}{1+x^2}}$$

i.e $1\le\sqrt{\int_0^1(1+x^2)f^2(x)dx}.\frac{\sqrt{\pi}}{2}$

$\therefore$ Required minimum is $\frac{4}{\pi}$ which is attained for $f(x)=\frac{4}{\pi(1+x^2)}$

59. **2**

For $\frac{4}{\pi}\sin^{-1}(a^8+1)+2\cos^{-1}(a^{12}+1)-\sec^{-1}(a^2+1)$ to be defined a = 0

$\therefore 3x-4y=\frac{4}{\pi}.\frac{\pi}{2}+2.0-0=2$

$3x-4y=2$

$6x-5y=7$

$d=\left|\frac{\frac{7}{2}-2}{5}\right|=\frac{3}{10}$

$r=\frac{3}{20}$

$l=r\theta=\frac{3}{20}\times\theta=2$

60. **1**

$t^2f(x)-2tf'(x)+f''(x)=0$ has equal roots

$D=0\Rightarrow\frac{f''(x)}{f'(x)}=\frac{f'(x)}{f(x)}$

$\Rightarrow \ln(f'(x))=\ln(f(x))-\ln C$

$\Rightarrow f(x)=Cf'(x)$

$\Rightarrow f(0)=C.f'(0)$

$\Rightarrow C=\frac{1}{2}$

$\therefore \frac{f'(x)}{f(x)}=2\Rightarrow \ln f(x)=2x+k$

$\Rightarrow \ln(f(0))=k\Rightarrow k=0$

$\Rightarrow \ln(f(x))=2x\Rightarrow f(x)=e^{2x}$

$t^2e^{2x}-4te^{2x}+4e^{2x}=0$

$\lim_{x\to0}\left(\frac{f(x)-1}{x}-\frac{t}{2}\right)=\lim_{x\to0}\left(\frac{e^{2x}-1}{2x}\times2-\frac{2}{2}\right)=1$